SpaceX Falcon 9

First Stage Landing Prediction

IBM Applied Data Science Capstone — Final Report

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github.com/hamzasalmahi-lab/ibm-data-science-capstone-spacex

Outline

Executive Summary

Methods

- Data collection from SpaceX REST API + Wikipedia web scraping
- Data wrangling — outcome label engineering (Class 0/1)
- Exploratory data analysis with Pandas & SQL
- Interactive analytics — Folium maps & Plotly Dash app
- Predictive modelling — four classifiers with GridSearchCV

Key Results

88.9%

Decision Tree test accuracy — best model

- KSC LC-39A has the highest site success rate
- Success rate has risen from 0% (2010) to ~84% (2020)
- ES-L1, GEO, HEO, SSO orbits show 100% landing success
- Heavier payloads correlate with lower landing success

Introduction

Background

SpaceX advertises a Falcon 9 launch at roughly $62M, while comparable launches from other providers cost upwards of $165M. The cost advantage comes from reusing the first stage. When the first stage lands successfully, the booster can be recovered and refurbished; when it does not, the launch effectively absorbs the full hardware cost.

Problem Statement

Given the publicly available launch record (site, payload mass, orbit, booster version, flight number and other features), can we predict whether a Falcon 9 first stage will land successfully? A reliable prediction would let a competitor — or an analyst pricing a bid — estimate the true marginal cost of any given launch.

Data

SpaceX REST API + Wikipedia (Falcon 9 launches)

Target

Landing outcome — Class 0 (failed) / Class 1 (landed)

Value

Per-launch cost estimate for bidding & analysis

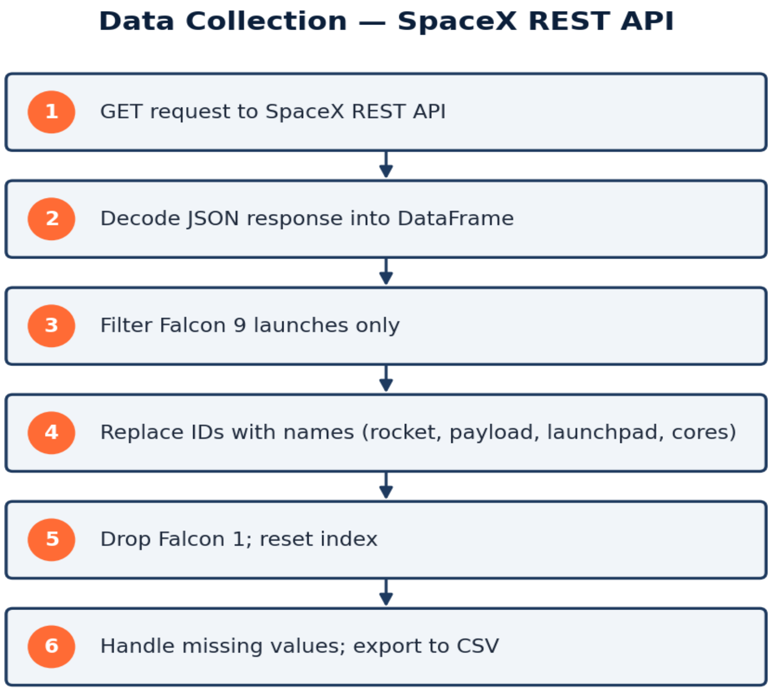
Methodology

End-to-end pipeline used throughout this project

1. **Data Collection** — SpaceX API + Wikipedia scraping
2. **Data Wrangling** — Cleaning + outcome label
3. **EDA — Viz & SQL** — Pandas + Plotly + SQL queries
4. **Interactive Analytics** — Folium + Plotly Dash
5. **Predictive Analysis** — 4 classifiers + GridSearchCV

Each stage is implemented in its own Jupyter notebook in the project repository, with the Dash app as a standalone Python file.

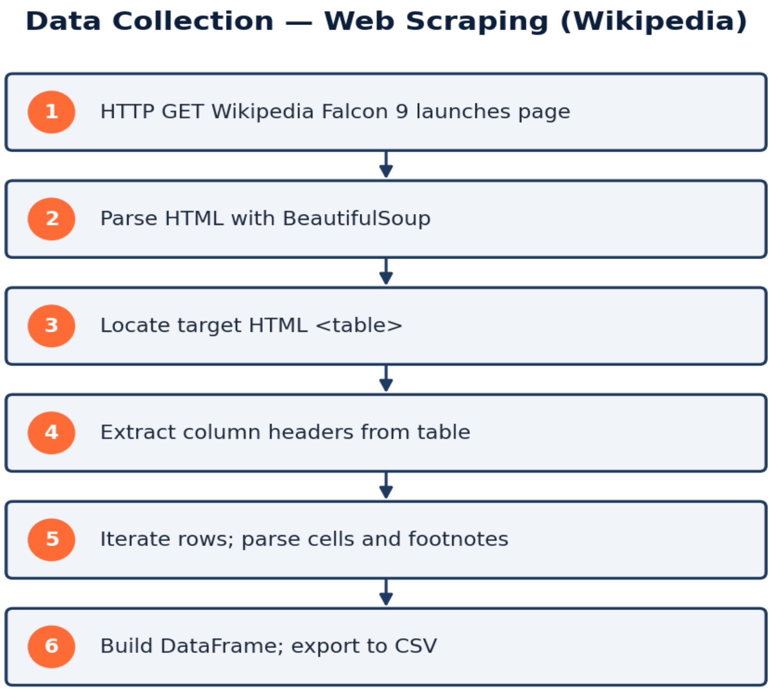
Data Collection — SpaceX REST API



Key elements

- Endpoint: api.spacexdata.com/v4/launches/past
- Tooling: requests + pandas + json_normalize
- Filter: Falcon 9 only (drop Falcon 1)
- Joins: 4 helper API calls to resolve rocket, payload, launchpad, core IDs
- Date filter: launches up to 13 Nov 2020
- Output: dataset_part_1.csv (90 rows × 17 columns)

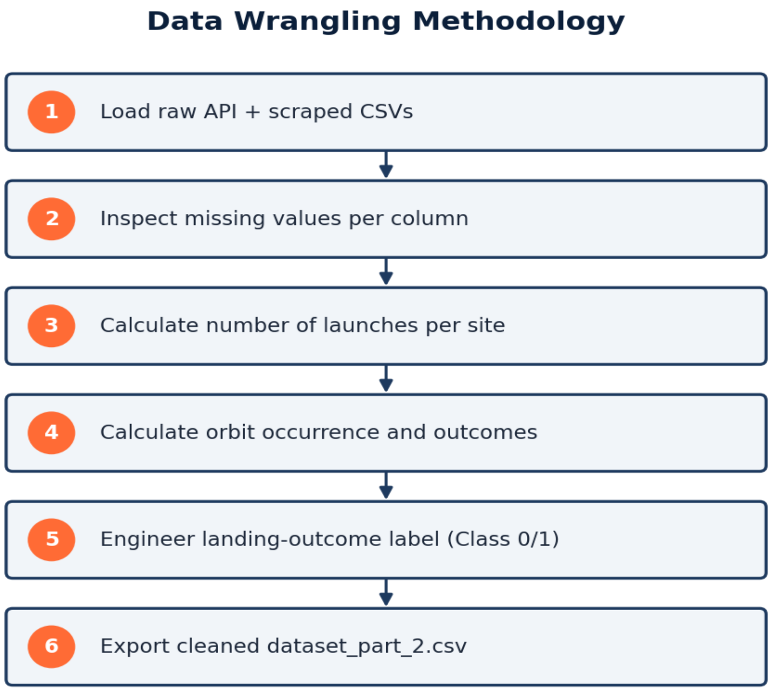
Data Collection — Web Scraping (Wikipedia)



Key elements

- Source: en.wikipedia.org — List of Falcon 9 and Falcon Heavy launches
- Tooling: requests + BeautifulSoup4
- Target: HTML table of historical launches
- Custom helpers: date_time(), booster_version(), landing_status()
- Output: spacex_web_scraped.csv (~121 rows × 11 columns)

Data Wrangling Methodology



Key elements

- Identify nulls per column; impute PayloadMass with column mean; drop LandingPad
- Count launches per Site and Orbit type
- Map Outcome strings into binary Class label (0/1)
- bad_outcomes = {None None, False ASDS, False Ocean, False RTLS, None ASDS}
- Mean overall landing success ≈ 67%
- Output: dataset_part_2.csv

EDA with Data Visualization

Charts used and what they reveal

Flight Number vs. Launch Site

Scatter — does the site matter as SpaceX gained flight experience?

Payload Mass vs. Launch Site

Scatter — which sites take heavier payloads, and at what success rate?

Success Rate vs. Orbit Type

Bar — which orbits land successfully most often?

Flight Number vs. Orbit Type

Scatter — are some orbits a recent capability?

Payload Mass vs. Orbit Type

Scatter — how does payload distribute across orbits?

Yearly Launch Success Trend

Line — improvement of landing success across 2010 – 2020

EDA with SQL

Ten queries executed against the cleaned dataset in SQLite

1. Display all unique launch sites
2. Show 5 records where launch site begins with 'CCA'
3. Total payload mass carried by boosters from NASA (CRS)
4. Average payload mass carried by F9 v1.1 boosters
5. Date of the first successful landing on ground pad
6. Booster names with successful drone-ship landings (4000 – 6000 kg)
7. Total number of successful vs. failed mission outcomes
8. Booster versions that carried the maximum payload
9. Records of failed drone-ship landings in 2015 (month, booster, site)
10. Rank landing outcomes between 2010-06-04 and 2017-03-20

Interactive Visual Analytics

Folium — Launch Site Maps

- Markers for all four SpaceX launch sites
- MarkerCluster of successes/failures per site
- Coloured icons by landing outcome
- Proximity analysis — coastline, railway, highway and nearest city
- Implementation: folium.Map, Marker, Circle, MarkerCluster, MousePosition, PolyLine

Plotly Dash — Launch Records Dashboard

- Dropdown to filter launch site (All / per site)
- Range slider for payload mass (0 – 10 000 kg)
- Pie chart — total successful launches share
- Scatter — payload vs. landing outcome, coloured by booster version
- Implementation: Dash, dash-html, dash-core, plotly.express

Predictive Analysis — Methodology

1 Build features

Numerical features + one-hot encoded categoricals

2 Standardise + split

StandardScaler · train_test_split (80 / 20)

3 Train 4 classifiers

Logistic Regression · SVM · Decision Tree · KNN

4 Tune hyperparameters

GridSearchCV with 10-fold cross-validation

5 Evaluate

Test accuracy + confusion matrix per model

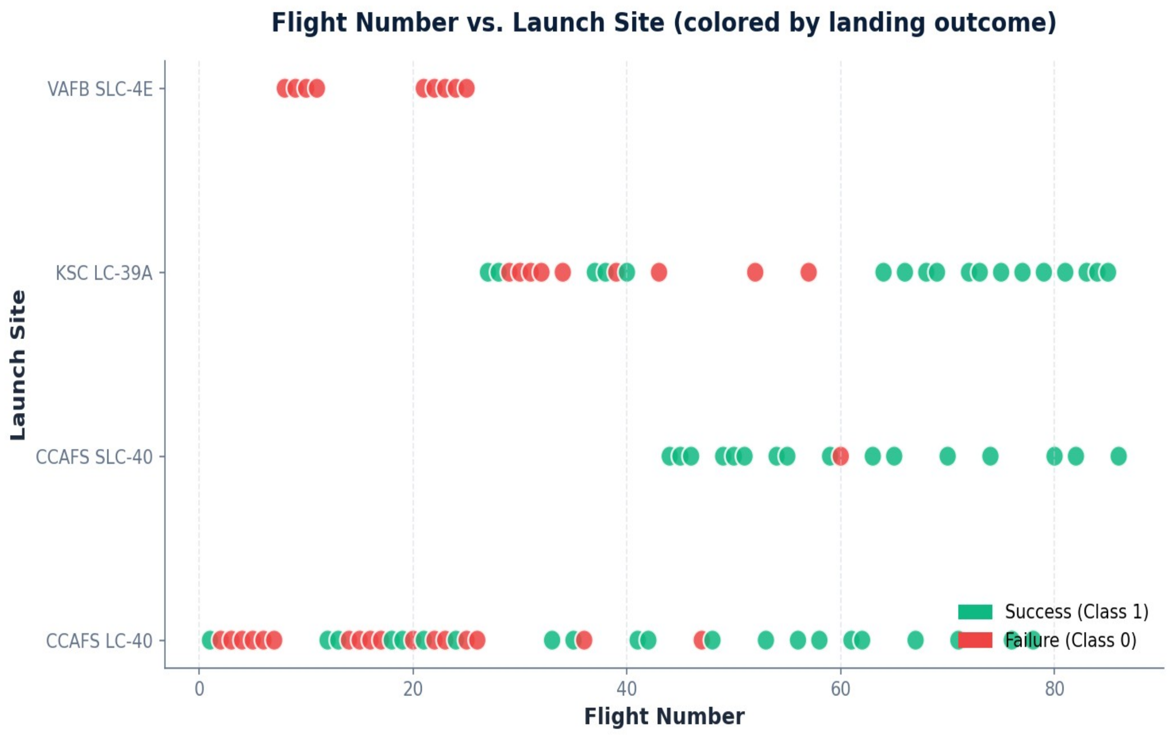
6 Select best model

Highest test accuracy ⇒ chosen model

Results

EDA visualisations · SQL queries · Interactive analytics · Predictive models

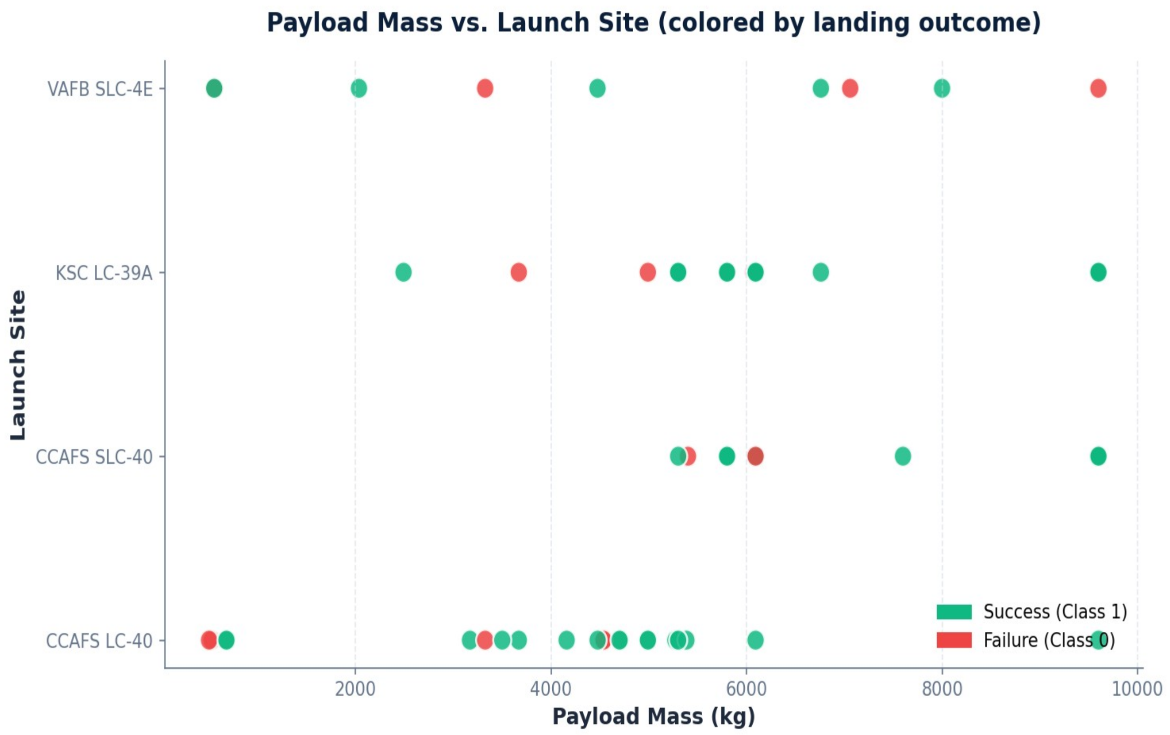
Flight Number vs. Launch Site



Insights

- Early flights cluster heavily at CCAFS LC-40
- KSC LC-39A enters service later (~flight 27)
- Success rate at each site improves as flight number grows
- VAFB SLC-4E sees relatively few launches

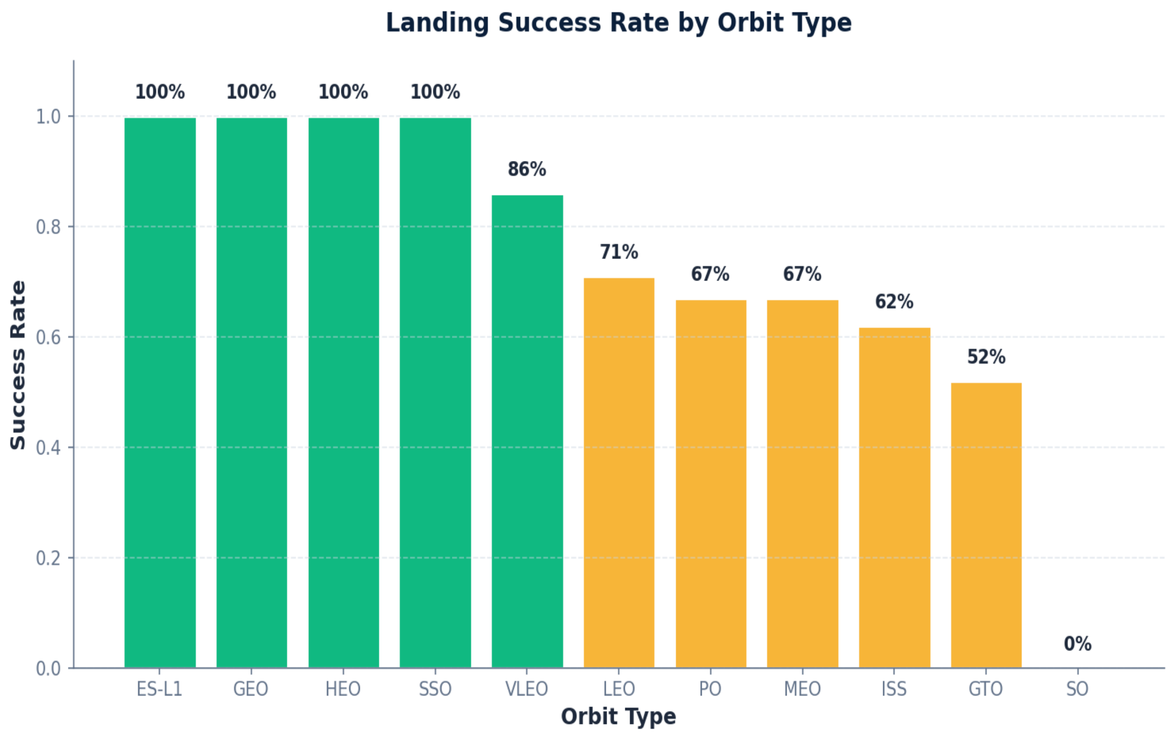
Payload Mass vs. Launch Site



Insights

- VAFB SLC-4E rarely flies > 10 000 kg payloads
- Heavy payloads cluster at CCAFS / KSC sites
- Higher payload mass is associated with reduced landing success at every site

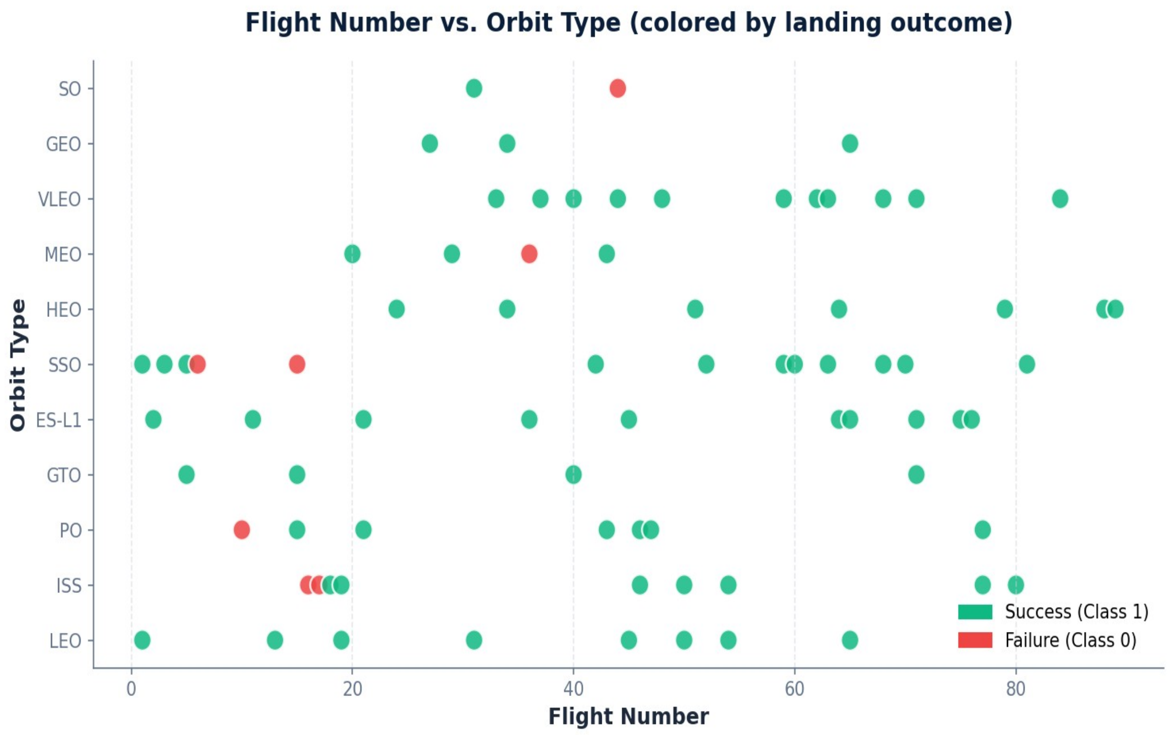
Landing Success Rate by Orbit Type



Insights

- Orbits with 100% success: ES-L1, GEO, HEO, SSO
- GTO and SO are the most challenging — lowest success
- LEO/ISS are mid-range, dominated by Dragon resupply missions

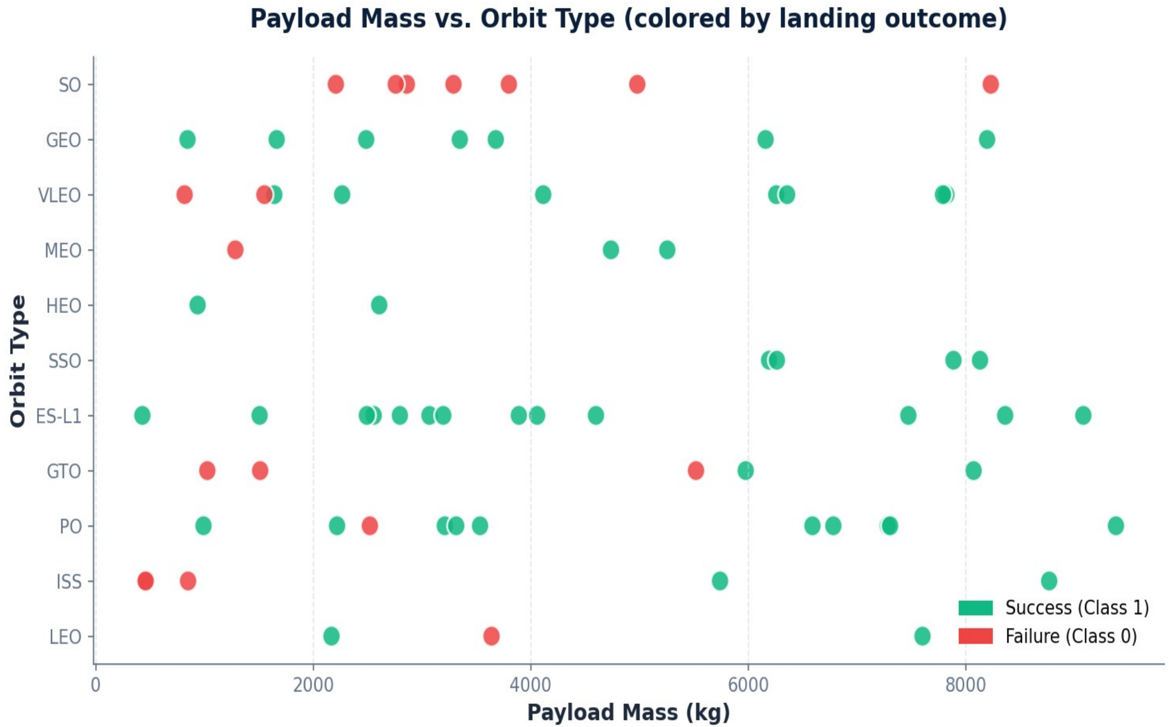
Flight Number vs. Orbit Type



Insights

- LEO and ISS missions span the full flight history
- VLEO is a newer capability — recent flight numbers only
- GTO success rate improves with flight experience

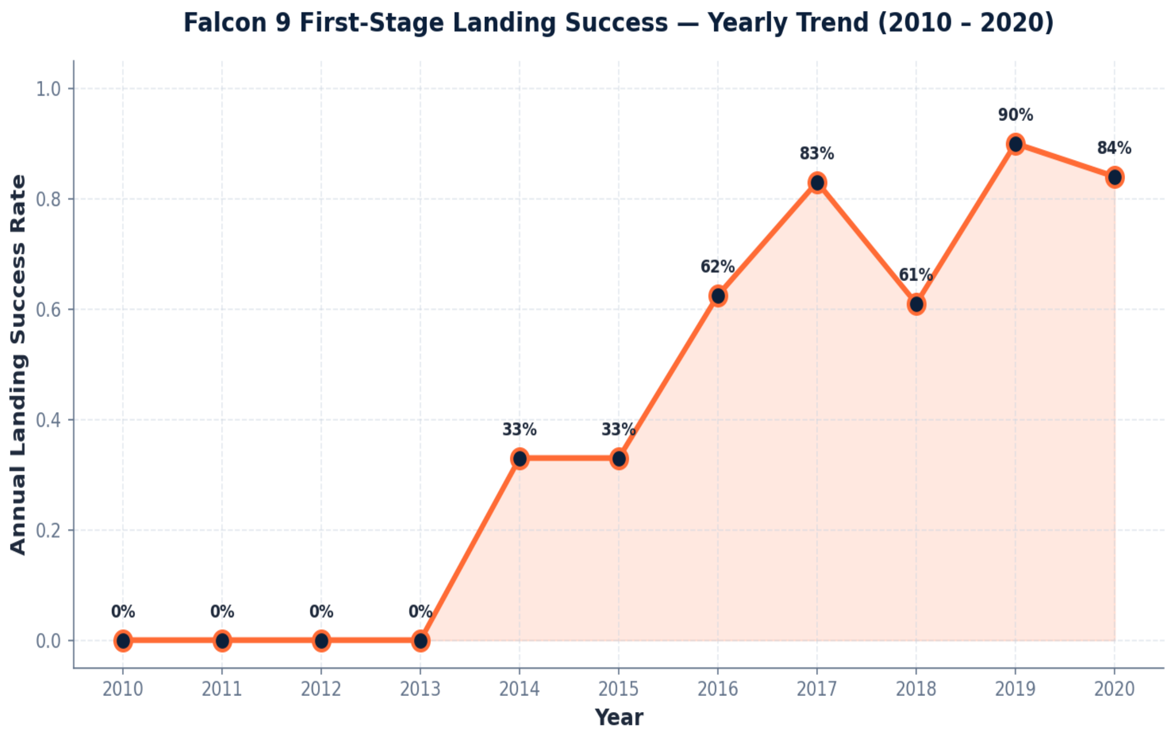
Payload Mass vs. Orbit Type



Insights

- Heavy payloads (> 9 000 kg) concentrated in LEO and VLEO
- Sub-GTO inserts mostly carry payloads under 8 000 kg
- Each orbit shows a characteristic payload range

Launch Success — Yearly Trend



Insights

- First successful landing was achieved in 2014
- Success climbs from 0% (2010 – 13) to ~84% (2020)
- Brief dip in 2018 (Block-4 attrition), recovers in 2019
- Flight number / year are strong predictive features

SQL — Unique Launch Sites

Query

```
SELECT DISTINCT Launch_Site
FROM   SPACEXTBL;
```

Result

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Takeaway

Four distinct launch sites — three on the Florida coast and one at Vandenberg, CA.

SQL — 5 Records with Launch Site Starting 'CCA'

Query

```
SELECT *  FROM SPACEXTBL
WHERE  Launch_Site LIKE 'CCA%'
LIMIT  5;
```

Result

Date	Booster_Version	Launch_Site	Payload	Mission_Outcome
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	Success
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1	Success
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2+	Success
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	Success
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	Success

Takeaway

All early CCAFS LC-40 missions succeeded at the mission level (booster landings come later).

SQL — Total Payload Mass (NASA CRS)

Query

```
SELECT SUM(PAYLOAD_MASS__KG_) AS total_payload_kg
FROM   SPACEXTBL
WHERE  Customer = 'NASA (CRS)';
```

Result

total_payload_kg
45 596

Takeaway

Across all CRS resupply launches, ~45.6 t of payload was lifted for NASA.

SQL — Average Payload Mass (F9 v1.1)

Query

```
SELECT AVG(PAYLOAD_MASS__KG_) AS avg_payload_kg
FROM   SPACEXTBL
WHERE  Booster_Version = 'F9 v1.1';
```

Result

avg_payload_kg
2 928.4

Takeaway

Average payload for F9 v1.1 boosters is ~2.93 t — a smaller earlier-generation vehicle.

SQL — First Successful Ground-Pad Landing

Query

```
SELECT MIN(Date) AS first_success_date
FROM   SPACEXTBL
WHERE  Landing_Outcome = 'Success (ground pad)';
```

Result

first_success_date
2015-12-22

Takeaway

First successful ground-pad recovery — 22 Dec 2015 (ORBCOMM OG-2 mission, LZ-1).

SQL — Drone-Ship Landings, Payload 4 000 – 6 000 kg

Query

```
SELECT Booster_Version
FROM   SPACEXTBL
WHERE  Landing_Outcome = 'Success (drone ship)'
  AND  PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

Result

Booster_Version
F9 FT B1022
F9 FT B1026
F9 B4 B1021.2
F9 B5 B1031.2

Takeaway

Four boosters meet both filters — all in the FT / B4 / B5 generations.

SQL — Total Successful vs. Failed Mission Outcomes

Query

```
SELECT Mission_Outcome, COUNT(*) AS n
FROM   SPACEXTBL
GROUP  BY Mission_Outcome;
```

Result

Mission_Outcome	n
Success	99
Success (payload status unclear)	1
Failure (in flight)	1

Takeaway

Mission success ≠ landing success — almost every mission succeeds at the payload-delivery level.

SQL — Booster Versions Carrying the Maximum Payload

Query

```
SELECT Booster_Version
FROM   SPACEXTBL
WHERE  PAYLOAD_MASS__KG_ = (
         SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL
       );
```

Result

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2

Takeaway

All maximum-payload (15 600 kg) boosters are F9 Block 5 vehicles.

SQL — 2015 Failed Drone-Ship Landings

```
Query
SELECT substr(Date,6,2) AS month,
       Booster_Version,
       Launch_Site
FROM   SPACEXTBL
WHERE  substr(Date,1,4) = '2015'
  AND  Landing_Outcome = 'Failure (drone ship)';
```

Result

month	Booster_Version	Launch_Site
01	F9 v1.1 B1012	CCAFS LC-40
04	F9 v1.1 B1015	CCAFS LC-40

Takeaway

Two failed drone-ship attempts in 2015, both with F9 v1.1 from CCAFS LC-40.

SQL — Landing Outcomes 2010-06-04 to 2017-03-20

Query

```
SELECT Landing_Outcome, COUNT(*) AS n
FROM   SPACEXTBL
WHERE  Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP  BY Landing_Outcome
ORDER  BY n DESC;
```

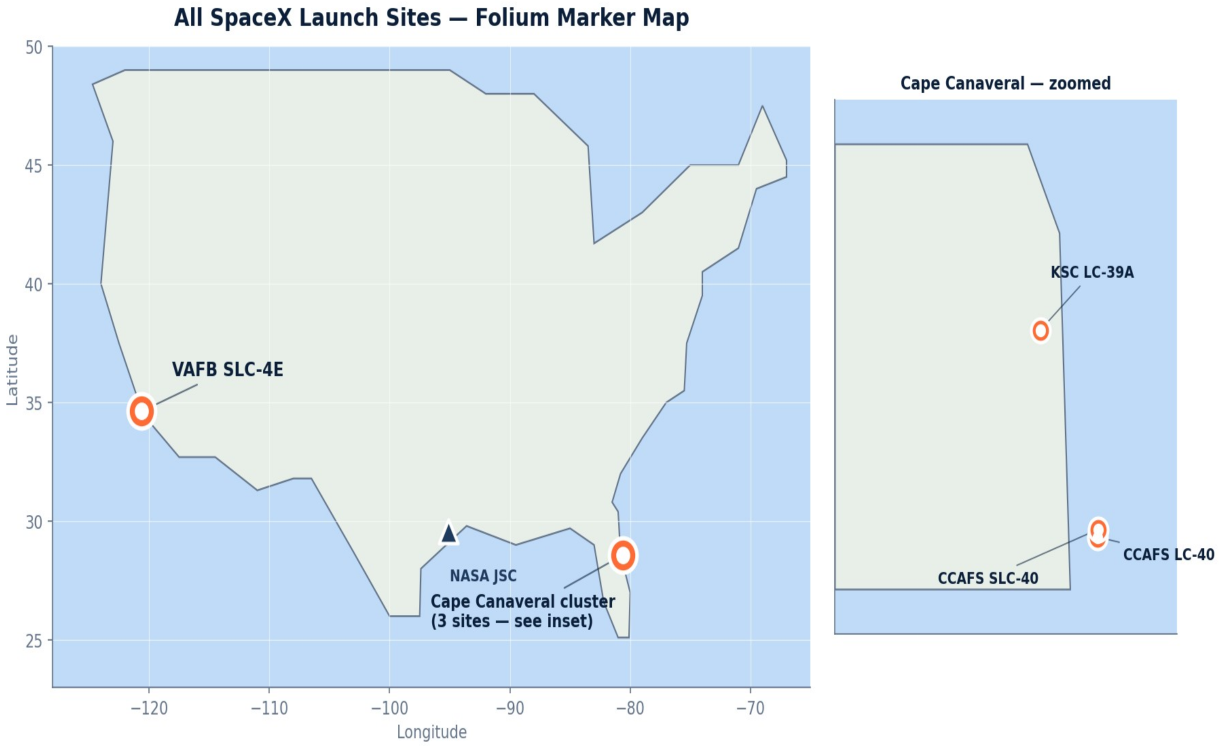
Result

Landing_Outcome	n
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Takeaway

'No attempt' is the most common outcome in this window — recovery only matured later in the dataset.

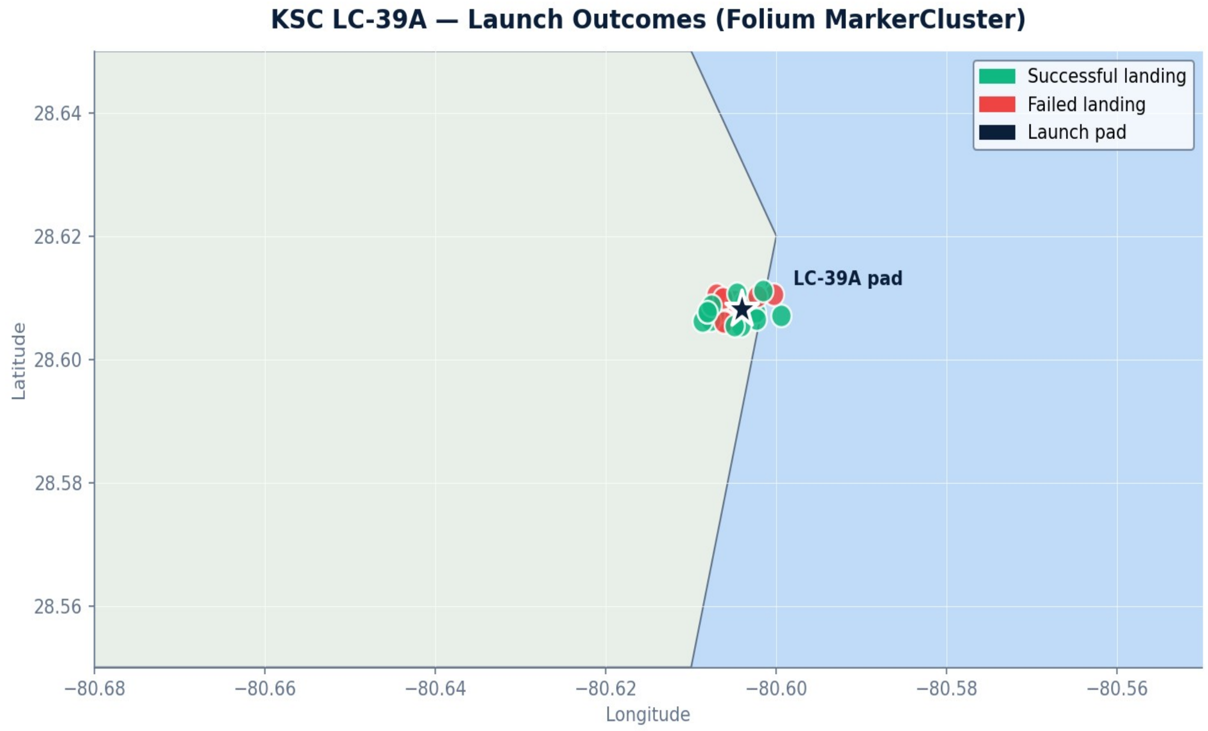
Folium — All Launch Site Markers



Observations

- All US launch sites lie close to the coast (east or west)
- Three Florida sites cluster within ~5 km — inset zoomed
- VAFB SLC-4E is the only west-coast site, used for polar / SSO insertions
- Coastal positioning enables debris fall-back into water

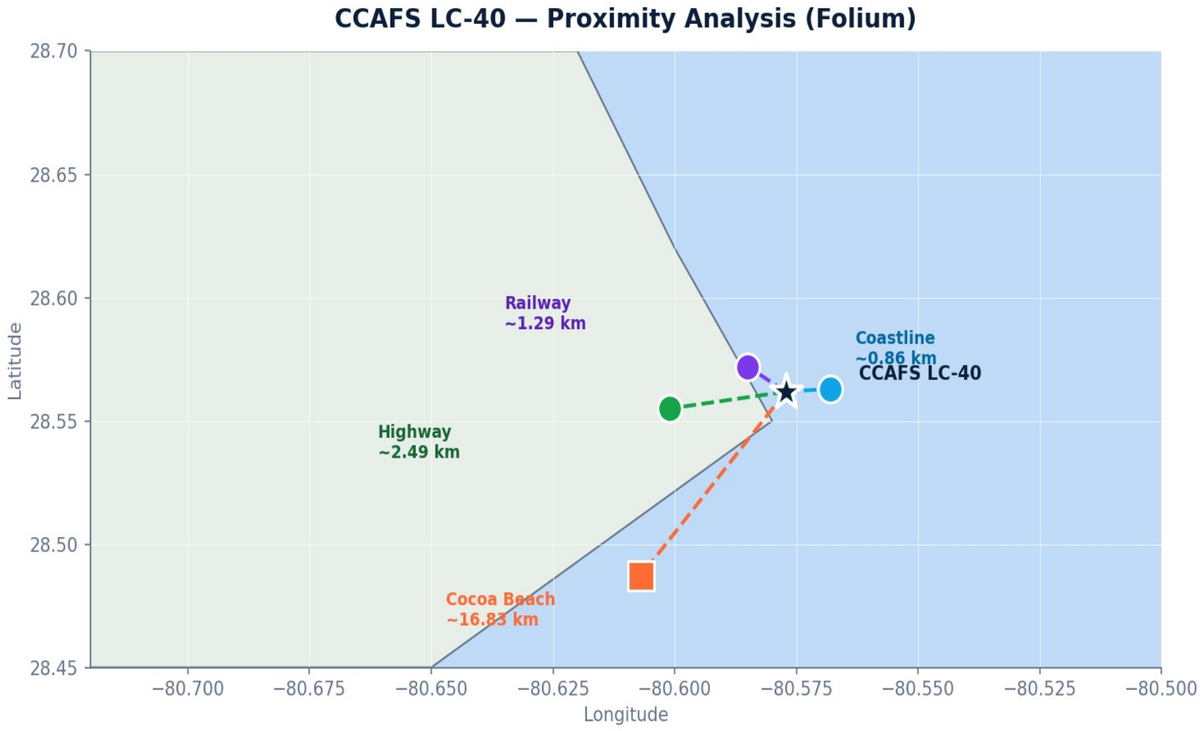
Folium — Launch Outcomes per Site



Observations

- Green markers = successful landings, red = failed
- Clustered at the LC-39A pad, then jittered for readability
- Successes outnumber failures by ~3 : 1 at this site
- MarkerCluster collapses to summary count when zoomed out

Folium — Proximity Analysis (CCAFS LC-40)

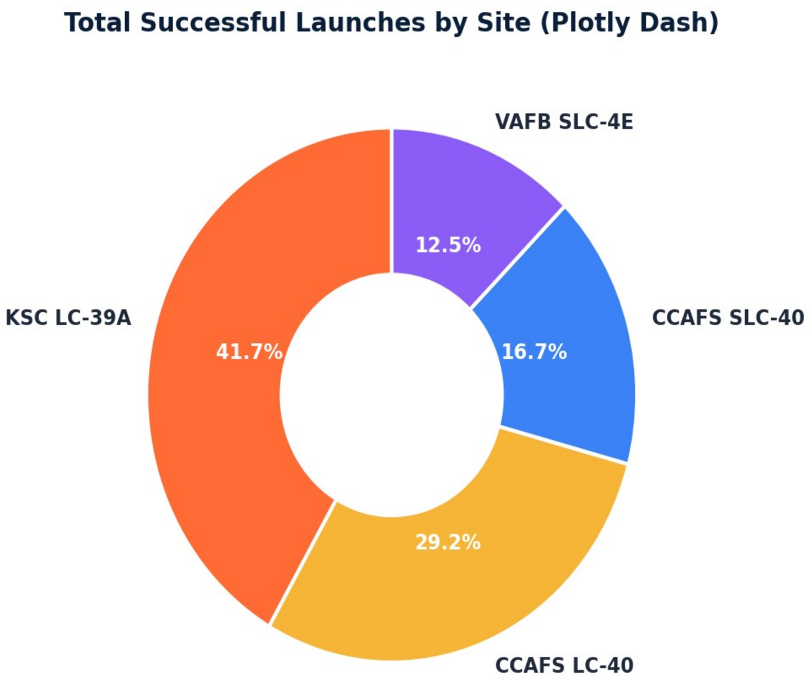


Distances

- Coastline: ≈ 0.86 km
- Railway: ≈ 1.29 km
- Highway: ≈ 2.49 km
- Cocoa Beach (nearest city): ≈ 16.83 km

Confirms launch sites stay near rail/road for logistics but far from population centres for safety.

Plotly Dash — Total Success by Site

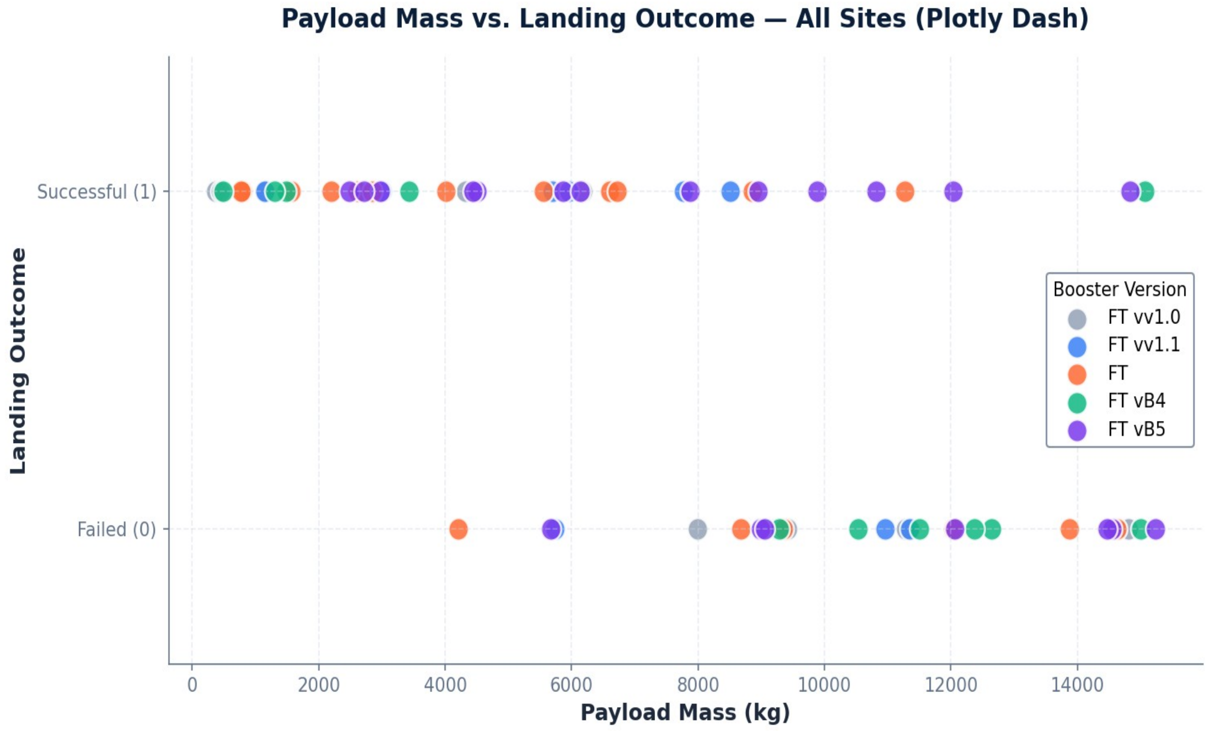


Pie chart — Insights

- KSC LC-39A contributes 41.7% of successful launches
- CCAFS LC-40 (29.2%) is the second-largest contributor
- VAFB SLC-4E has the smallest share (12.5%)
- Selecting a single site replaces the pie with that site's success-vs-failure breakdown

[Dash app — spacex_dash_app.py]

Plotly Dash — Payload vs. Landing Outcome

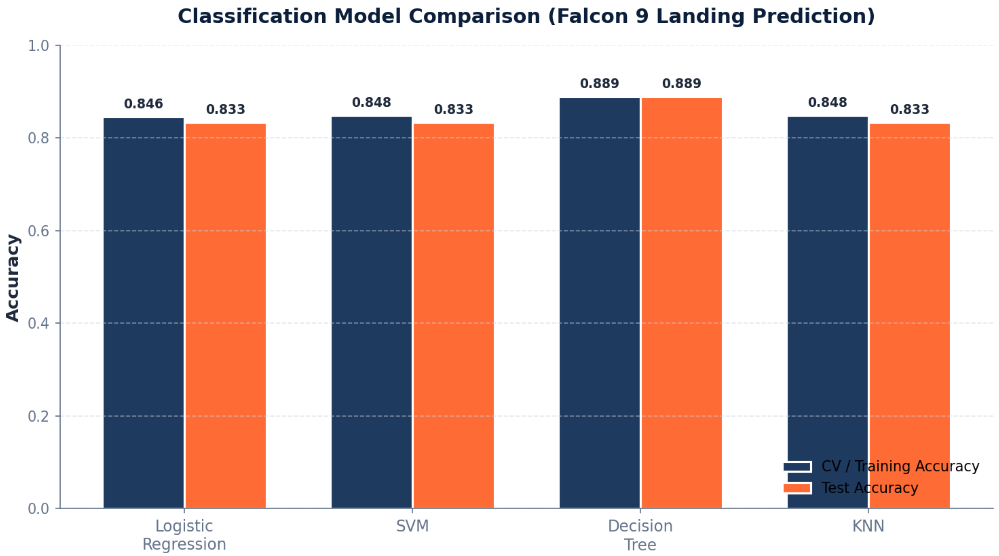


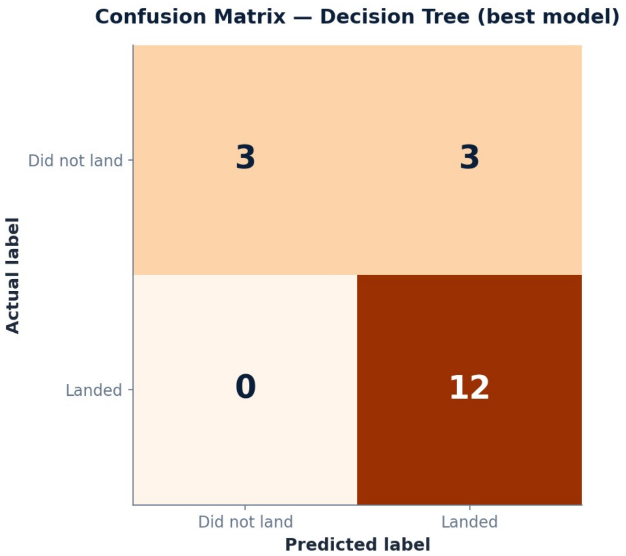
Insights

- Higher payload mass → lower success probability
- Block 5 boosters dominate the high-payload, successful region
- v1.0 / v1.1 confined to low-mass payloads
- Slider on the dashboard lets the analyst restrict the payload window dynamically

Dash app — spacex_dash_app.py

Predictive Analysis — Model Comparison & Confusion Matrix





Decision Tree — best model: 88.9% test accuracy

CM (test n=18): 12 true positives (Landed), 3 true negatives (Did not land), 0 false negatives, 3 false positives — recall on the 'Landed' class is 100%.

Conclusions & Insights

Best model

Decision Tree

88.9% test accuracy

- Recall (Landed) = 100%
- Recall (Did not land) = 50%
- Precision (Landed) = 80%

Useful for predicting which launches will land successfully — but the model is more conservative about predicting failures.

What we learned

- Launch site matters: KSC LC-39A has the highest success rate; VAFB the lowest volume
- Orbit matters: ES-L1, GEO, HEO and SSO show 100% landing success; GTO/SO are hardest
- Flight number is itself a feature: success rises monotonically with experience
- Payload mass and landing success are inversely related — denser payloads cost more recoveries
- Tree-based models capture non-linear interactions (orbit × payload × site) that linear models miss
- Creative extension: the same pipeline can estimate per-launch marginal cost, useful for any bidder pricing against SpaceX

Thank you.

Questions are welcome.

Project repository

github.com/hamzasalmahi-lab/ibm-data-science-capstone-spacex

Hamza S. Almahi, MBBS · Cairo, Egypt · June 2026